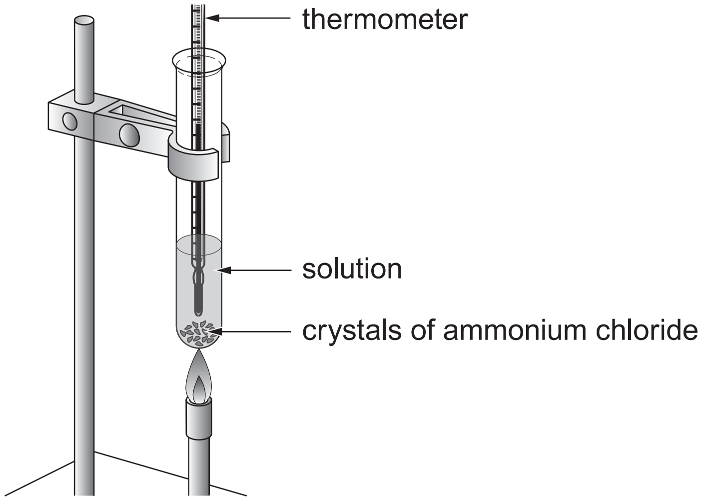
Examiner only

4. A student investigates the solubility of ammonium chloride by adding different masses to 10 g of water.

He uses the apparatus shown.



10 g of water is placed in a boiling tube and 3.0 g of ammonium chloride is added.

The tube is heated until all the solid dissolves.

The tube is allowed to cool.

The temperature at which solid ammonium chloride first appears is recorded.

The experiment is repeated using different masses of ammonium chloride.

The results are shown in the table.

Mass of ammonium chloride in 10 g of water (g)	3.0	3.3	4.1	5.2	5.9	6.6
Temperature at which solid ammonium chloride first appears (°C)	4	10	30	52	68	80

(a) What practical problem is the student likely to come across in finding the first two results? Suggest how this problem might be overcome. [2]

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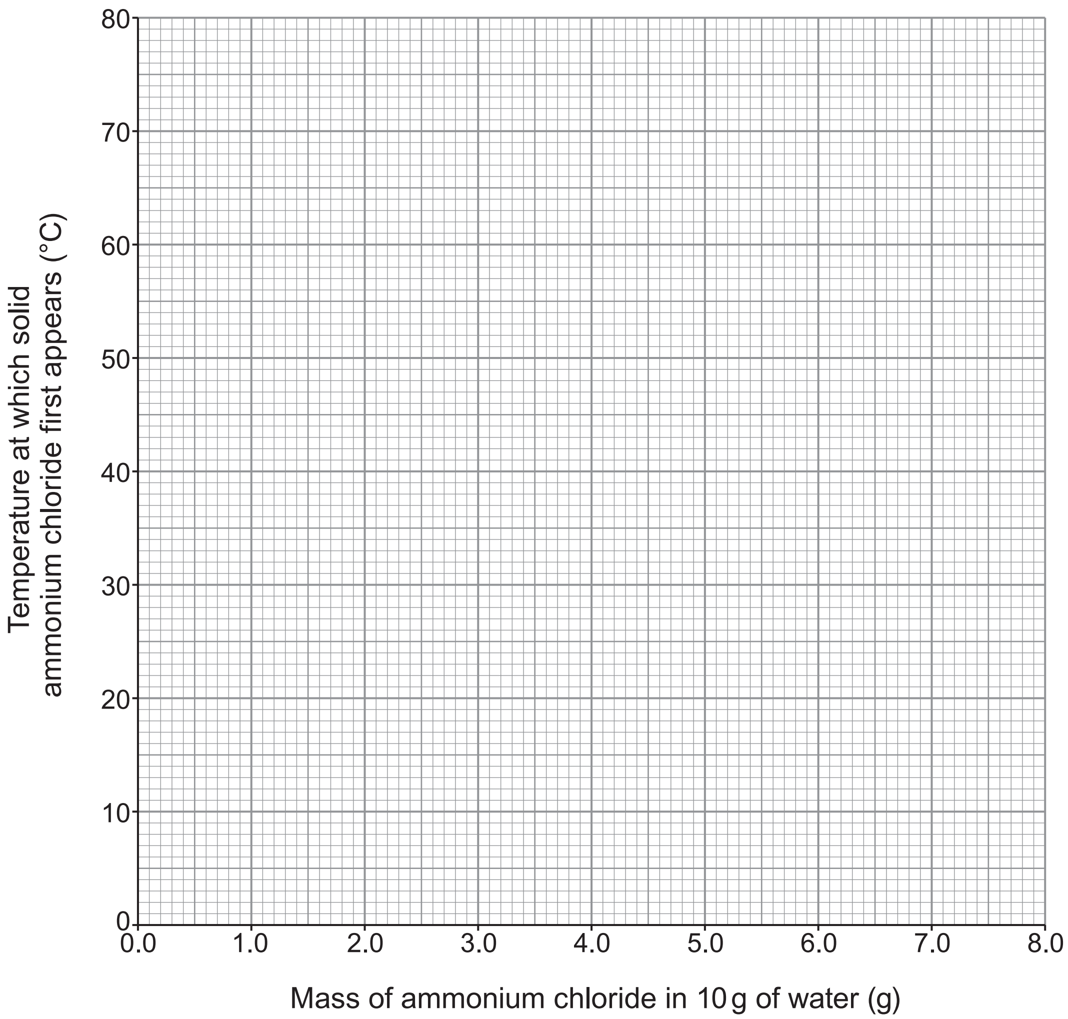
(b) (i) On the grid below, plot the temperature at which solid ammonium chloride first appears against the mass of ammonium chloride in 10 g of water. Draw a suitable line. [3]



(ii) The student is given a boiling tube containing 5.0 g of ammonium chloride in 10 g of water. He stirs the ammonium chloride in the water and heats it to a temperature of 45 °C.

State whether all the ammonium chloride dissolves. Give a reason for your answer. [1]

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(c) The student is asked to use a **different** method to find the exact solubility of another compound in water at room temperature. He knows that it has a value of approximately 7 g per 100 g of water at this temperature.

He is given a 5.0 g sample of the compound and common laboratory equipment but **no heating apparatus**.

Describe how he would carry out his method and how he would find the solubility. [3]

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